

Pranav Rathod

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EDUCATION

University of Southern California

Master of Science in Computer Science - Multimedia and Creative Technologies

Los Angeles, CA

GPA: 3.68/4.00

University of Illinois at Chicago

Bachelor of Science in Computer Science, Minor in Art

Chicago, IL

GPA: 3.92/4.00

RELEVANT COURSEWORK

Multimedia Systems Design, Analysis of Algorithms, Computer Graphics, Computer Animation and Simulation, Game Engine Development, 3D Graphics and Rendering, Augmented/Virtual Reality, Artificial Intelligence, Framework-based Development

SKILLS

Languages: C/C++, Java, Python, C#, Dart, Go, SQL, F#, JavaScript

Technical: OpenGL, PyTorch, WebGL, Unity, Vuforia Engine, Blender, Flutter SDK, GCP, Firebase, SQLite3, Multi-Threading, Locking, TCP/IP, Android Studio, Google/JUnit5 Testing Framework, Docker, Git, MacOS, Windows, Unix/Linux,

EXPERIENCE

Media Support Specialist

January 2024 – Present

Annenberg TechOps Information Technology, University of Southern California

Los Angeles, CA

- Leveraging web and design expertise and providing coding support to make custom news feed and article pages.
- Answering technical questions and provided assistance with HTML, Adobe Creative Cloud and WordPress.
- Producing screen casts and step-by-step tutorials to aid users in multimedia production.

Research Assistant

June 2022 – December 2023

ELiCIT Lab, University of Illinois at Chicago

Chicago, IL

- Delivered Flutter app for Natural-User Interaction studies, utilizing GCP's Speech-to-Text, Meta's Detectron2 and PyTorch.
- Investigated distributed system design trade-offs and caching strategies during app development and backend optimization.
- Collaborated with a multidisciplinary team of 5 researchers to integrate user feedback into UI enhancements.
- Explored existing research on creative technologies for children, contributing to a comprehensive literature review.

Teaching Assistant

January 2022 – December 2022

Department of Computer Science, University of Illinois at Chicago

Chicago, IL

- Guided students in debugging and writing code in C, C++ and building applications using Flutter, Dart across 3 semesters.
- Formulated course logistics and graded assignments for intermediate and advanced Computer Science courses each semester.
- Utilized communication platforms such as Piazza and Blackboard to effectively instruct more than 100 students at once.

PROJECTS

IK System with Skinning | C++, OpenGL, Eigen, ADOL-C

April 2025

- Developed a character deformation system combining linear blend skinning, forward kinematics, and inverse kinematics.
- Implemented joint hierarchy traversal to compute local/global transforms and skinning matrices in real time.
- Used Eigen for IK system solving and ADOL-C for automatic Jacobian computation, enabling smooth, stable handle manipulation with Tikhonov regularization.

Segmented Video Compression | Python, Detectron2, OpenCV, PyTorch

December 2024

- Designed a compression pipeline to encode .rgb videos using DCT, quantization, and motion estimation.
- Advanced segmentation by using Detectron2 and optical flow, achieving 95% accuracy in foreground-background separation.
- Constructed a video player enabling 30 FPS playback with real-time IDCT reconstruction, and synchronized audio.

Roller Coaster Simulation | C++, OpenGL

March 2024

- Engineered a roller coaster simulation with Catmull-Rom splines in OpenGL, using control points from a custom format file.
- Elevated appearance with rail cross-section rendering and introduced a texture-mapped ground with Phong shading.
- Automated frame capture for animation with an immersive first-person view and realistic camera movement.

Virtual Reality Kiosk | Unity, C#, Virtual Reality ToolKit (VRTK), Blender

October 2023

- Developed a Virtual Reality environment showcasing the interior of a proposed college building.
- Allowed users to move and interact with objects by deploying application in a VR headset.
- Built custom 3D models leveraging Blender and mapped custom textures drawn using Procreate.

ACHIEVEMENTS & LEADERSHIP

SIGGRAPH Student Volunteer Team Leader | Denver, CO; Tokyo, Japan; Vancouver, BC

2024–2025

- Oversaw operations for 350+ international student volunteers at world's premier computer graphics conference
- Facilitated event logistics and attendee support, collaborating with global teams of students and industry professionals

Undergraduate Research Forum | Chicago, IL

April 2023

- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research

Engineering Expo | Chicago, IL

April 2021

- Won 'Best in Category' for designing an Arduino-based device to enforce COVID-19 social distancing